

Letter of Submittal

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Steven Winiski
Manager, Dometic Vancouver
Dometic Corporation
3831 No. 6 Road
Richmond, BC

Dear Mr. Winiski,

Please accept this technical report summarizing the ignition SCADA transition project I worked on during my Spring 2026 Co-op term at Dometic Vancouver. These projects focus on further improving, optimizing, and operationalizing the SCADA Asset Management System and creating the Air Filtration System for warehouse air barriers.

This report outlines the motivation behind both projects, the methodologies used to expand and operationalize the SCADA-based maintenance system and develop the air filtration retrofit, the technical challenges encountered during implementation, and the resulting improvements to maintenance operations and facility infrastructure at Dometic Vancouver. The work was conducted at the Dometic Vancouver facility in Richmond, BC, located on the traditional and unceded territories of the Musqueam, Squamish, and Tsleil-Waututh Nations.

The report is original and has not been submitted for credit elsewhere. I worked under your supervision as my manager, and collaborated closely with Process Development Engineer Huseyin Doganay the SCADA project lead, and Operations Engineering Manager Rob Pehl, who provided guidance on machine and bench naming, work cell structures, and PM validation.

Sincerely,

Ryan Thomas

Operational Expansion of SCADA-Based Asset Management and Air Filtration Retrofit
Design at Dometic Vancouver

Technical Report - Spring 2026 Co-op Term

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Abstract

This report summarizes two major engineering initiatives completed during my co-op term at Dometic Vancouver: the operational expansion of the facility's SCADA-based asset management system (AMS) and the iterative design, fabrication, and implementation of an air filtration retrofit for warehouse air barriers. The SCADA initiative focused on transitioning the newly established maintenance tracking and scheduling hub from a developmental framework into an actively deployed operational system. This included finalizing asset nomenclature tagging and hierarchy organization for machines/workbenches, uploading and structuring large volumes of preventive maintenance (PM) data, launching automated weekly PM scheduling and notification workflows, and generating monthly maintenance performance reports for management. Since deployment in January, the system has maintained preventive maintenance completion rates exceeding 85% over rolling biweekly windows, demonstrating strong adoption and validating the platform's effectiveness in digitizing maintenance operations.

The second project addressed oil contamination issues affecting warehouse air barriers caused by airborne condensed oil entering intake vents in manufacturing areas and resulting in oil leaks and slipping hazards. In an attempt to eliminate this issue, I designed a modular sheet metal air filtration system then fabricated, prototyped, and iteratively improved through field testing. Following prototype installation and several months of testing, I refined the design to improve performance, accessibility for workers, and better accommodate environmental constraints. Preliminary airflow analysis performed in Creo indicates approximately 5–8% airflow leakage in the final design while maintaining exceptional ventilation performance.

Together, these projects improved maintenance visibility, operational reliability, environmental cleanliness, and long-term facility infrastructure management at Dometic Vancouver while further developing skills in system design, automation, CAD, fabrication, technical communication, and engineering problem solving.

Introduction

Preventive maintenance is essential in a manufacturing environment where machines/workbenches directly support production output and safety. At Dometic Vancouver, PM activities have historically been recorded using the outdated methods of paper sheets and multiple partially overlapping spreadsheets

During my co-op term, I contributed to two engineering projects aimed at improving operational efficiency and facility reliability across the organization. The first involved expanding and operationalizing the AMS I helped develop during my previous co-op term. While the original system architecture and foundational asset data had been established, the platform required significant additional development, data population, workflow refinement, and organizational rollout before it could function as a fully integrated maintenance management system.

The second project addressed persistent oil contamination issues affecting warehouse air barriers. Condensed oil generated by nearby machines was being drawn into the air barrier intake systems, resulting in oil leakage and increased maintenance requirements. Air barrier configurations lacked adequate filtration and were constrained by challenging installation environments involving nearby piping and limited access clearances.

The purpose of this report is to document the engineering work completed on both projects, including the technical challenges addressed, methodologies employed, results achieved, and recommendations for future development.

SCADA Asset Management System Expansion

Problem Identification and Objectives

My objective for this semester was to get the AMS up and running. Significant work remained to transform the platform from a structured data repository into an automated functioning maintenance management tool.

Key outstanding limitations included incomplete preventive maintenance task integration, unfinished scheduler configuration, inconsistent legacy data formatting, and the absence of live automated maintenance dispatching. Without these capabilities, maintenance activities continued to rely heavily on fragmented manual coordination and legacy scheduling methods.

The primary objectives of this semester:

1. Finalize tagging, labeling, and organization of all machines/workbenches within the AMS asset hierarchy.
2. Upload, filter, and structure preventive maintenance task data for integration into the scheduler.
3. Activate automated PM scheduling and notification workflows.
4. Establish maintenance logging and reporting systems for long-term performance tracking.
5. Support system development and future scalability planning.

Methodology and Implementation

The first major task involved finalizing the digital asset registry for the facility. Existing machine/workbench data were reviewed, filtered, and standardized to ensure consistency in naming conventions, descriptions, and hierarchical placement within the AMS. Legacy records often contained outdated, duplicated, or inconsistent naming structures, requiring significant manual review and verification through consultation with facility personnel and direct observation on the production floor.

Following asset finalization, preventive maintenance data from numerous legacy spreadsheets and maintenance documents were reviewed and prepared for upload into the AMS scheduler. This process required extensive filtering and restructuring of maintenance records to ensure PM frequencies, personnel, dates, instructions, and asset associations were properly configured for automated scheduling.

Once sufficient PM data had been prepared, weekly PM scheduling workflows were activated. Under the implemented workflow, the maintenance manager activates PM tasks, then a custom script in Ignition automatically dispatches weekly maintenance task emails every Thursday to designated personnel. Each task is issued with a 14 day completion window.

Upon completion of assigned maintenance, personnel updates the maintenance logs, generating a permanent maintenance history within the system. This enables supervisors and engineers to monitor maintenance completion trends, overdue tasks, and historical servicing records in real time.

In addition to data management responsibilities, I became increasingly involved in system administration and development support. I assisted users with system-related questions, handled requests for new machine/bench additions, attended project meetings as a key representative for AMS, and collaborated with the Process Engineer to observe and support development-side logic implementation within Ignition.

Results and Discussion

Since operational deployment began in January, the AMS platform has demonstrated strong adoption and measurable success. The system has consistently achieved preventive maintenance completion rates exceeding 85% over biweekly completion windows, indicating successful integration into departmental workflows.

Monthly and year-to-date maintenance performance reports are now generated using exported system data, categorized by operator, maintenance, and outsourced contractor tasks. These reports are compiled into professional graphical summaries and distributed to the Director of Operations, Operations Manager, and Maintenance Manager to support decision-making and accountability.

Beyond the quantitative performance improvements, the project has significantly improved maintenance visibility, accountability, and long-term data accessibility. Maintenance history is now centrally stored, searchable, and continuously updated, eliminating reliance on fragmented spreadsheets and paper records.

Challenges encountered during implementation primarily stemmed from legacy data inconsistencies and the complexity of configuring logic within a live operational automation environment. Naming ambiguities for machines/benches required repeated consultation and verification, while development-side scripting challenges required iterative debugging and trial/error troubleshooting. Despite these obstacles, the system has progressed from conceptual deployment into active operational use and is now functioning as a scalable maintenance platform.

Future expansion plans include integrating facility infrastructure assets such as piping, ventilation systems, washrooms, and other building systems into the platform, with the long-term goal of establishing a centralized operational hub for all Dometic Vancouver asset management.

Air Filtration/Ventillation Design for Warehouse Air Barriers

Problem Identification and objectives

Air barriers throughout Dometic Vancouver were experiencing oil leakage and contamination due to airborne condensed oil entering intake systems from nearby machines. This contamination created cleanliness concerns, increased maintenance requirements, and slipping hazards throughout the warehouse.

Complicating the problem, many air barriers were installed in physically constrained environments near piping, walls, and other obstructions, limiting the design for retrofit solutions.

The objectives of this project were to:

1. Design a modular filtration retrofit to prevent oil contamination within air barriers.
2. Ensure the design accommodated environmental and spatial constraints.
3. Maintain acceptable airflow and air barrier performance.
4. Optimize the design for manufacturability and ease of filter replacement.
5. Fabrication across all eight air barrier units.

Design Methodology and Iterative Development

The initial design phase in my first semester focused on developing a custom sheet metal ventilation/filter housing that could mount directly over the air barrier intakes while fitting within the physical constraints of each installation environment. Site measurements were taken to account for nearby obstructions and available installation clearances.

To improve maintainability, replaceable filter media positioned over the intake opening, the design prioritized straightforward filter access and replacement due to the frequency of filter replacement

Fabrication of the sheet metal housing components was outsourced for cutting, bending, and welding. Once fabricated, custom 3D-printed inserts designed in-house were installed into the housings and secured using screws to create the final filter retention mechanism.

I installed and tested the prototype in the field several months. Testing demonstrated extremely minimal oil leakage, validating the filtration concept; however, observations indicated the initial housing geometry was oversized relative to the selected filter media, creating unnecessary width and poor airflow.

Based on my observations, I redesigned the vent for a smaller filter, allowing the housing to be reduced in size and better aligned with the width of the air barrier itself. The filter insertion/removal mechanism was also redesigned to allow single-side filter replacement, improving serviceability and simplifying maintenance. This redesign required creation of left and right specific housing configurations.

To validate the revised geometry, airflow analysis was performed in Creo. Preliminary simulation results indicate the updated design introduces approximately 5–8% airflow leakage, which significantly improves filtration effectiveness and reduces contamination risk.

Results and Discussion

The original ventilation system successfully mitigates the primary contamination issue while maintaining acceptable airflow performance and improving maintainability. Prototype testing confirmed substantial reduction in oil ingress compared to the unfiltered configuration, with only minimal leakage observed during extended field testing.

The iterative redesign even further reduced the oil leakage as the design directed all the surrounding air straight through the filter into the intake, while also addressing fit and accessibility issues with the first design

Based on current performance, the estimated filter replacement interval is approximately three to four months, though this frequency may increase following the second iteration and long-term operational monitoring.

With the design now nearing finalization, fabrication of eight production units is planned for installation across all warehouse air barriers.

Conclusion and Recommendations

This co-op term resulted in successful completion of two major engineering initiatives that provided measurable operational and facility improvements at Dometic Vancouver.

The AMS project successfully transitioned from developmental infrastructure into a live operational maintenance management platform. Through completion of asset organization, PM integration, scheduler deployment, automated notifications, reporting workflows, and system administration support, the platform is now actively managing maintenance operations with completion rates exceeding 85%. The project has significantly improved maintenance visibility, accountability, and long-term data management capabilities while establishing a scalable foundation for future facility-wide asset integration.

The air filtration retrofit project successfully developed and validated a modular engineered solution for mitigating oil contamination in air barriers. Through iterative design, prototyping, field testing, and airflow validation, a manufacturable and scalable filtration system was created that significantly reduces oil ingress while maintaining acceptable airflow performance.

Key recommendations moving forward include:

1. Continue expanding AMS asset coverage to include facility infrastructure and utilities.
2. Refine PM scheduling logic to better accommodate maintenance department workload variability.
3. Monitor AMS performance metrics over longer operational periods to identify scheduling optimization opportunities.
4. Deploy all eight air filtration retrofit units and track long-term filter life and contamination reduction performance.
5. Observe the new ventilation iteration and make improvements accordingly.

Overall, these projects strengthened Dometic Vancouver's maintenance automation, facility reliability, and infrastructure management capabilities while providing valuable practical experience in engineering design, automation systems, manufacturing coordination, and technical problem solving.